

Chat Insights

The best questions and takeaways from the Session 4 live chat

Session 4 drew one of the busiest chats of the cohort so far, with the room splitting its energy between logistics (a meetup, an AI Day, and a schedule debate), sharp technical questions on chunking and LangChain, and a genuinely strong back and forth on RAG categories and vector search internals. This document pulls out the questions and answers most learners will want to revisit, attributed to whoever asked or answered them in the chat.

1. Community and Schedule

Before the technical content began, the chat carried a lively discussion about two upcoming events and a recurring question about session timing.

Asked by	Question	Answer or takeaway
Nitesh	Any plan for a meetup in Hyderabad or Noida?	A Codebasics meetup is set for September 19 (Saturday) near Princeton, New Jersey, for the East Coast crowd. A separate in person event, Codebasics AI Day, runs September 5 (Teacher's Day) in Hyderabad, 5 to 8:30 PM
Bharath Kumar	Is class suspended on September 5, since it clashes with AI Day?	No. Co instructors already announced at inauguration will run that class, so there is no clash for anyone staying in the regular session
Karthikeyan M S	Could the session time move by two hours (6 to 9 PM) to suit more time zones?	Sid ran an informal show of hands live in the session. A clear majority preferred to keep the current time, several citing that this timing was a condition of enrollment, so no change is planned

Key Insight

Ashish clarified in chat that any future change to session timing will always be announced through Discord and email, so there is no need to track rumors or DMs about it.

2. LangChain: Why Use It in Production

Asked by	Question	Answer or takeaway
Anush Narayanan	Yesterday we built RAG without LangChain using the same embedding model and LLM. What does LangChain actually add?	Syed Mizbahuddin and Dhaval Patel answered together: you can swap the vector database, the LLM provider, or the embedding model by changing a library instead of rewriting the pipeline, which avoids vendor lock in, and it does the same work in far fewer lines of code
amit s	In production grade applications, do teams use LangChain or pure Python?	Not directly answered live, but the surrounding discussion pointed to LangChain being common where teams value speed of development and provider flexibility over full manual control
Nltesh	Is it correct that LangChain cannot store metadata?	Ashish[Team Codebasics] corrected this in chat: LangChain does support metadata, and shared the LangChain documentation on knowledge bases for reference
Renganathan M	If RAG is one use case, what else does LangChain offer?	Ashish[Team Codebasics]: building agents, tool calling, and prompt management, among others, pointing to LangChain's overview documentation
Amit Patil	If I download a model along with its tokenizer and embedding logic, can I run a LangChain pipeline fully offline?	Syed Mizbahuddin confirmed he was doing exactly that, though noted that Colab disconnecting means re downloading the embedding model periodically

3. Chunking, Tokens, and Overlap

Asked by	Question	Answer or takeaway
Murthy Rajender	What is the right chunk overlap, and how do we decide it?	Syed Mizbahuddin and Ashish[Team Codebasics]: there is no single right answer, it is experimental. A common starting point is 15 to 20 percent of the chunk size, evaluated against real retrieval quality
Abhisekh	What is the benefit of creating tokens before chunking, instead of chunking directly on characters or words?	Sid and several learners (akhil, Sanjeev Rohila) converged on the same point: chunking by raw characters can split a word or token in half, while tokenizing first respects the model's own boundaries and keeps embedding and API costs predictable
Ashu Swami	How do you decide which chunking strategy to use across a large set of documents you cannot inspect individually?	Sid: build a generic pipeline that parses documents by their actual layout, summarizes any images or diagrams, then applies hierarchical or hybrid chunking. No pipeline will be 100 percent robust, but this covers the large majority of documents
supreet kaur	What is the difference between hierarchical and layout aware chunking?	Layout aware chunking is hierarchical chunking extended to also understand tables, columns, and diagrams on the page, using a vision model, rather than assuming the text reads top to bottom in a single column
Sharath Sheshadri	Are custom chunking and hybrid chunking the same thing?	Ashish[Team Codebasics]: no. Hybrid chunking can be custom, but custom chunking does not necessarily mean hybrid, it depends on the use case, the documents, and the retrieval requirement
Abdoul	What chunking strategy works best for 100 pages of mathematical content, with formulas, equations, and annexes?	Not answered live in the transcript, flagged here as worth raising again since it did not get a direct response

Standout Participant Insight (from akhil)

"AI models measure text size in tokens, not words or characters. Chunking with a tokenizer ensures each piece fits the model's maximum limit... APIs charge money based on token counts, so tokenizer chunking helps you track and predict your exact usage costs." A clean summary of why the tokenizer step exists at all.

4. RAG Categories and Vector Search Internals

Asked by	Question	Answer or takeaway
Sandip	How is Graph RAG different from PageIndex (reasoning based RAG)?	Sid, from the session: Graph RAG builds and traverses a knowledge graph of entities and relationships. PageIndex does not build a knowledge graph at all, it only builds a hierarchical index of a document's sections, closer to a book's index than a web of connections
Mugdha	Vector RAG already stores keywords in its payload. How is that different from hybrid RAG?	Not directly answered live. Worth clarifying that hybrid RAG runs a dedicated keyword search algorithm (BM25 or TF IDF) alongside vector search and merges both result sets, rather than relying on keywords stored as payload metadata
Amit Patil	In hybrid RAG, could vector and keyword search return overlapping chunks and duplicate the context?	Flagged as a good question that stayed open in the chat, worth asking again when hybrid RAG hands on happens next week
renjith r	In keyword RAG, do we skip creating embeddings entirely and just use a database like Elasticsearch?	anuragp added the detail live: yes, this style of retrieval works off an index, such as the one BM25 builds, not off embeddings
Evgeni, Anush Narayanan	Will agentic RAG be covered in the cohort?	Sid confirmed yes, later in the cohort. He described it briefly as RAG where the retriever is handed to the LLM as one of its tools, instead of being called directly inside a fixed pipeline
Anush Narayanan	Is the traversal complexity of HNSW $O(\log N)$, similar to a binary search, and do vector databases handle this automatically when you build an HNSW index?	Confirmed correct in the surrounding discussion, and Syed Mizbahuddin confirmed Qdrant handles HNSW internally, so users do not need to implement it themselves
supreet kaur, Ravee	How exactly are HNSW's layers created, and how is it decided which chunk goes into which layer?	Sid: a chunk is placed based on its cosine similarity to nearby points already in the structure. Similar points cluster together, dissimilar points end up farther apart, and layers build up from there

Standout Participant Insight (from Saquib Siddiqui)

On visualizing HNSW's layers: "Consider layers as something like express lanes, where you are searching for a particular number within a large set of numbers." Paired well with Pawan's analogy in the same thread, comparing the search to traveling by flight for long distances, then bus or train, then car for the shortest stretch.

5. Tools and Resources Shared in Chat

Shared by	Resource
Dhaval Patel (Team Codebasics)	LinkedIn post announcing the Codebasics AI activity and meetup
sunil	codebasics.io/lp/codebasics-ai-day , the Hyderabad AI Day event page
Maulik Shah	codebasics.io/events , where all upcoming events are announced
Ashish[Team Codebasics]	Indian Data Club (community partner) social channels, for learners looking for regular meetups
Syed Mizbahuddin	tiktokenizer.vercel.app , for experimenting with different tokenizers including <code>cl100k_base</code>
Sid [Team Codebasics]	docs.unstructured.io and the LangChain document loader integrations page, as alternatives or companions to Docling
Ashish[Team Codebasics]	LangChain's knowledge base documentation (for metadata) and its overview documentation (for agents, tools, and prompt management)
raghav bansal	A PDF data extraction benchmark comparing document parsing tools (procycons.com)
Dhaval Patel (Team Codebasics)	A YouTube walkthrough of the custom chunking approach AtliQ uses internally